COMPUTER SCIENCE + COMPUTER ENGINEERING FOR GAMES

07

PACKAGE 8 · ASSESSMENT

Assessments

60 original core questions · 40 new transfer questions

Undergraduate computer-engineering core

Modules 31-35 · Gate G

Edition: September 28, 2026

A guided self-study curriculum. Supplied examples are learning references, not completed learner projects. Simulation is the default; physical hardware is optional.

Contents

Page references and PDF bookmarks are linked for navigation.

Instructions and scoring 3
Lesson 31.1 - Voltage, current, resistance and safe measurement 4
Lesson 31.2 - Capacitors, inductors, transients and frequency response 5
Lesson 31.3 - Diodes, transistors, CMOS and analog interfaces 6
Lesson 31.4 - Sampling, ADCs, DACs and noisy controller inputs 7
Lesson 32.1 - Boolean logic and combinational building blocks 8
Lesson 32.2 - Registers, finite-state machines and synchronous timing 9
Lesson 32.3 - Binary arithmetic, fixed point and overflow policies 10
Lesson 32.4 - Clock domains, reset and metastability 11
Lesson 33.1 - RTL semantics and synthesizable SystemVerilog 12
Lesson 33.2 - Testbenches, scoreboards and constrained stimulus 13
Lesson 33.3 - Assertions, coverage and formal reasoning 14
Lesson 33.4 - Synthesis, constraints and FPGA resource mapping 15
Lesson 34.1 - Instruction sets, datapaths and single-cycle execution 16
Lesson 34.2 - Pipelining, forwarding, stalls and branch recovery 17
Lesson 34.3 - Memory interfaces, caches and protection boundaries 18
Lesson 34.4 - Processor verification, toolchains and architectural evidence 19
Lesson 35.1 - Memory-mapped I/O, timers, interrupts and DMA 20
Lesson 35.2 - Framebuffers, scanout and pixel pipelines 21
Lesson 35.3 - A small accelerator and its software contract 22
Lesson 35.4 - Integrated digital-design review and optional board bring-up 23

Instructions and scoring

There are 100 questions: 60 unchanged core questions and 40 new transfer questions. IDs ending C1-C3 are original core questions; T1-T2 are added application questions. Each question is worth two points.

Attempt answers before opening the key. For the core quiz, two points requires a correct answer with the reasoning or units appropriate to the question; one point is substantially correct but incomplete; zero is missing or incorrect. Target at least 5/6 per lesson. Transfer questions form a separate /4 diagnostic, with a suggested target of 3/4. They do not replace the original core requirement.

Record explanations in a separate notebook or in the browser edition. Passing a quiz alone does not pass a lesson: the assigned reading and practical assignment must also be complete. Keep first and corrected attempts so that improvements are visible.

Lesson 31.1 - Voltage, current, resistance and safe measurement

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

31.1-C1 - 2 points. What current flows through 1,000 ohms at 5 V?

Answer / reasoning: _____

31.1-C2 - 2 points. What power is dissipated in that ideal resistor?

Answer / reasoning: _____

31.1-C3 - 2 points. Why can putting a meter in the wrong current-measurement configuration be dangerous?

Answer / reasoning: _____

31.1-T1 - 2 points. A 5 V source drives R1=2 kohm above R2=3 kohm. A 6 kohm load is placed across R2. Find Vout and all three branch currents.

Answer / reasoning: _____

31.1-T2 - 2 points. Why does testing 1,000 random resistor values inside a tolerance band not prove the worst-case divider voltage?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 31.2 - Capacitors, inductors, transients and frequency response

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

31.2-C1 - 2 points. What is tau for 10 kohm and 1 microfarad?

Answer / reasoning: _____

31.2-C2 - 2 points. What fraction of a zero-to-final RC charging step is reached at one tau?

Answer / reasoning: _____

31.2-C3 - 2 points. Why do initial conditions matter?

Answer / reasoning: _____

31.2-T1 - 2 points. For the 10 kohm, 1 microfarad low-pass, find the 90% rise time and cutoff frequency.

Answer / reasoning: _____

31.2-T2 - 2 points. Why is a forward-Euler RC response that overshoots at dt=1.5 tau not evidence that the ideal physical RC circuit overshoots?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 31.3 - Diodes, transistors, CMOS and analog interfaces

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

31.3-C1 - 2 points. Why is a transistor not only a binary software switch?

Answer / reasoning: _____

31.3-C2 - 2 points. Does a logical high guarantee compatibility across voltage domains?

Answer / reasoning: _____

31.3-C3 - 2 points. Why distinguish absolute maximum ratings from recommended operating conditions?

Answer / reasoning: _____

31.3-T1 - 2 points. A non-inverting ideal gain is 10 on a nominal 0-to-3.3 V supply. Does a 0.5 V input establish a valid 5 V output?

Answer / reasoning: _____

31.3-T2 - 2 points. Starting low, a comparator sets at >=1.8 V and clears at <=1.5 V. What outputs follow 1.4, 1.6, 1.81, 1.7, 1.49 V?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 31.4 - Sampling, ADCs, DACs and noisy controller inputs

First attempt: _____ Core: ___ / 6 Transfer: ___ / 4

31.4-C1 - 2 points. How many distinct codes does a 12-bit converter provide?

Answer / reasoning: _____

31.4-C2 - 2 points. Does 12-bit resolution guarantee 12-bit accuracy?

Answer / reasoning: _____

31.4-C3 - 2 points. Why filter before sampling to prevent aliasing?

Answer / reasoning: _____

31.4-T1 - 2 points. For alpha=0.25, how many discrete updates are needed to reach at least 90% of a unit step from zero?

Answer / reasoning: _____

31.4-T2 - 2 points. At fs=1 kHz, why can digital processing alone not identify whether an observed 300 Hz sampled cosine came from 300 or 700 Hz?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 32.1 - Boolean logic and combinational building blocks

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

32.1-C1 - 2 points. What distinguishes combinational from sequential logic?

Answer / reasoning: _____

32.1-C2 - 2 points. What does a two-to-one multiplexer do?

Answer / reasoning: _____

32.1-C3 - 2 points. Does the shortest Boolean expression always imply the fastest circuit?

Answer / reasoning: _____

32.1-T1 - 2 points. A four-bit adder receives unsigned 13 and 7. Give the low result, carry-out and number of exhaustive input cases including carry-in.

Answer / reasoning: _____

32.1-T2 - 2 points. Why does a one-hot assertion not fully verify a 2-to-4 decoder?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 32.2 - Registers, finite-state machines and synchronous timing

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

32.2-C1 - 2 points. What does a register capture in a synchronous model?

Answer / reasoning: _____

32.2-C2 - 2 points. What simplified maximum frequency follows from an 8 ns period?

Answer / reasoning: _____

32.2-C3 - 2 points. Does slowing the clock automatically fix hold violations?

Answer / reasoning: _____

32.2-T1 - 2 points. Given maximum delays 0.8+5.6+0.6 ns plus 0.5 ns uncertainty, what period is required and what is setup slack at 8 ns?

Answer / reasoning: _____

32.2-T2 - 2 points. A path has 0.15 ns earliest data arrival and a 0.25 ns hold requirement. What is its hold slack, and does changing period from 8 to 16 ns fix it?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 32.3 - Binary arithmetic, fixed point and overflow policies

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

32.3-C1 - 2 points. What is the signed 8-bit two's-complement range?

Answer / reasoning: _____

32.3-C2 - 2 points. What does stored value 24 mean with scale 1/16?

Answer / reasoning: _____

32.3-C3 - 2 points. Why use a wider intermediate for multiplication?

Answer / reasoning: _____

32.3-T1 - 2 points. Give wrapped and saturated signed eight-bit results for 120+20, then state the range of signed W=16,F=8.

Answer / reasoning: _____

32.3-T2 - 2 points. With nearest-even division by four, what do products +6, +10, -6 and -10 become?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 32.4 - Clock domains, reset and metastability

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

32.4-C1 - 2 points. Does a two-register synchronizer eliminate all metastability risk?

Answer / reasoning: _____

32.4-C2 - 2 points. Why not independently synchronize every bit of a changing binary bus?

Answer / reasoning: _____

32.4-C3 - 2 points. Can ordinary digital simulation prove an analog metastability failure rate?

Answer / reasoning: _____

32.4-T1 - 2 points. Why does a two-register synchronizer on a one-cycle pulse not guarantee that a slower destination sees the pulse?

Answer / reasoning: _____

32.4-T2 - 2 points. In a bundled-data handshake, when may the source change the payload, and what additional physical assumption is still required?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 33.1 - RTL semantics and synthesizable SystemVerilog

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

33.1-C1 - 2 points. Why assign every combinational output on every path?

Answer / reasoning: _____

33.1-C2 - 2 points. What does the nonblocking swap example do?

Answer / reasoning: _____

33.1-C3 - 2 points. Does a source-code loop necessarily take multiple hardware clock cycles?

Answer / reasoning: _____

33.1-T1 - 2 points. Before an edge a=1 and b=2. What follows a<=b; b<=a, and why?

Answer / reasoning: _____

33.1-T2 - 2 points. Why is an omitted else sometimes intentional in always_ff but dangerous in intended combinational logic?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 33.2 - Testbenches, scoreboards and constrained stimulus

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

33.2-C1 - 2 points. Should a scoreboard count attempted or accepted transactions?

Answer / reasoning: _____

33.2-C2 - 2 points. Why save random seeds?

Answer / reasoning: _____

33.2-C3 - 2 points. Why deliberately inject a known bug?

Answer / reasoning: _____

33.2-T1 - 2 points. Under this FIFO policy, depth two is full with [A,B], input offers C and output-ready is high. Which transfers occur and what remains?

Answer / reasoning: _____

33.2-T2 - 2 points. Why should the scoreboard use an abstract sequence instead of copying the RTL read/write-pointer equations?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 33.3 - Assertions, coverage and formal reasoning

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

33.3-C1 - 2 points. Does reaching 100% of chosen coverage bins prove correctness?

Answer / reasoning: _____

33.3-C2 - 2 points. Why can overconstrained assumptions hide a bug?

Answer / reasoning: _____

33.3-C3 - 2 points. What limitation must accompany a bounded proof?

Answer / reasoning: _____

33.3-T1 - 2 points. Does proving 0<=count<=DEPTH establish FIFO ordering?

Answer / reasoning: _____

33.3-T2 - 2 points. A bounded check finds no counterexample through 20 cycles. What claim is justified, and what must not be claimed?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 33.4 - Synthesis, constraints and FPGA resource mapping

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

33.4-C1 - 2 points. Does successful synthesis establish timing closure?

Answer / reasoning: _____

33.4-C2 - 2 points. Why inspect inferred memory resources?

Answer / reasoning: _____

33.4-C3 - 2 points. Why use identical constraints when comparing designs?

Answer / reasoning: _____

33.4-T1 - 2 points. At width 16 and depth eight, how many abstract payload bits are present, and how wide must occupancy be?

Answer / reasoning: _____

33.4-T2 - 2 points. Why cannot a generic synthesis cell count establish routed FPGA Fmax?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 34.1 - Instruction sets, datapaths and single-cycle execution

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

34.1-C1 - 2 points. How does an ISA differ from a microarchitecture?

Answer / reasoning: _____

34.1-C2 - 2 points. What limits a simple single-cycle clock period?

Answer / reasoning: _____

34.1-C3 - 2 points. Why avoid claiming full ISA compliance for a small subset?

Answer / reasoning: _____

34.1-T1 - 2 points. Why does G16 ADD r3,r1,r2 encode as 0x2650?

Answer / reasoning: _____

34.1-T2 - 2 points. What must happen when LD r0 uses an out-of-range address in the G16 contract?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 34.2 - Pipelining, forwarding, stalls and branch recovery

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

34.2-C1 - 2 points. Does pipelining always shorten one instruction's latency?

Answer / reasoning: _____

34.2-C2 - 2 points. Why must flushed instructions suppress writes?

Answer / reasoning: _____

34.2-C3 - 2 points. What is a load-use hazard?

Answer / reasoning: _____

34.2-T1 - 2 points. Why can forwarding the ALU result of a load fail to solve its immediate load-use dependency?

Answer / reasoning: _____

34.2-T2 - 2 points. A taken branch skips a store, but the final registers match the reference. Is that enough to validate branch recovery?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 34.3 - Memory interfaces, caches and protection boundaries

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

34.3-C1 - 2 points. Why is an instantaneous array an incomplete memory model?

Answer / reasoning: _____

34.3-C2 - 2 points. What does a cache tag identify?

Answer / reasoning: _____

34.3-C3 - 2 points. Does a small CPU necessarily need an MMU?

Answer / reasoning: _____

34.3-T1 - 2 points. A 32-bit byte address uses 16-byte blocks and 64 direct-mapped lines. Give offset, index and tag widths.

Answer / reasoning: _____

34.3-T2 - 2 points. A store request is accepted once, then its response is delayed five cycles. How many times may the memory write occur?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 34.4 - Processor verification, toolchains and architectural evidence

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

34.4-C1 - 2 points. Why compare at retirement rather than arbitrary internal cycles?

Answer / reasoning: _____

34.4-C2 - 2 points. Why inspect compiler-generated instructions?

Answer / reasoning: _____

34.4-C3 - 2 points. What belongs in a reproducible processor failure report?

Answer / reasoning: _____

34.4-T1 - 2 points. Why compare retirement events rather than register files after the same number of cycles in a reference interpreter and pipeline?

Answer / reasoning: _____

34.4-T2 - 2 points. Does generating random legal programs with the same decoder code used by the DUT provide fully independent verification?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 35.1 - Memory-mapped I/O, timers, interrupts and DMA

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

35.1-C1 - 2 points. Why are device registers not ordinary memory variables?

Answer / reasoning: _____

35.1-C2 - 2 points. When may software reuse a DMA source buffer?

Answer / reasoning: _____

35.1-C3 - 2 points. Does C++ volatile make a shared device driver thread-safe?

Answer / reasoning: _____

35.1-T1 - 2 points. With pending=1, clear=1 and event=1, what does set-wins W1C logic produce, and why?

Answer / reasoning: _____

35.1-T2 - 2 points. Can holding a device response valid for five cycles justify repeating the original enqueue five times?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 35.2 - Framebuffers, scanout and pixel pipelines

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

35.2-C1 - 2 points. How many bytes does the example framebuffer require?

Answer / reasoning: _____

35.2-C2 - 2 points. Is active-pixel bandwidth the whole physical interface bandwidth?

Answer / reasoning: _____

35.2-C3 - 2 points. Why synchronize framebuffer swaps?

Answer / reasoning: _____

35.2-T1 - 2 points. For 64x48 active pixels inside 80x60 total periods at 288 kHz, find frame frequency and active samples per frame.

Answer / reasoning: _____

35.2-T2 - 2 points. Why can delaying RGB by one register while leaving active unchanged corrupt an otherwise correct frame?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 35.3 - A small accelerator and its software contract

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

35.3-C1 - 2 points. What total time follows from halving the example 20 microsecond compute phase?

Answer / reasoning: _____

35.3-C2 - 2 points. What does backpressure accomplish?

Answer / reasoning: _____

35.3-C3 - 2 points. Why report latency and throughput separately?

Answer / reasoning: _____

35.3-T1 - 2 points. Software costs 2 us+8N ns and hardware 30 us+3N ns. What is the smallest integer N for a strict modeled hardware win?

Answer / reasoning: _____

35.3-T2 - 2 points. If 20% of application time speeds up 4x, what is the ideal total speedup before new overhead?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____

Lesson 35.4 - Integrated digital-design review and optional board bring-up

First attempt: _____ **Core:** ___ / 6 **Transfer:** ___ / 4

35.4-C1 - 2 points. Why are unit tests insufficient for integration?

Answer / reasoning: _____

35.4-C2 - 2 points. What must be checked before wiring a peripheral?

Answer / reasoning: _____

35.4-C3 - 2 points. Does a successful FPGA demonstration prove an ASIC is ready to fabricate?

Answer / reasoning: _____

35.4-T1 - 2 points. Can a sprite-compositor capstone replace the required Module 34 processor exercise?

Answer / reasoning: _____

35.4-T2 - 2 points. A generic synthesis run succeeds and a board later displays an image. What stronger claims still need separate evidence?

Answer / reasoning: _____

Correction or uncertainty to revisit: _____